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# Transpose and Flatten ★

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## Transpose

We can generate the transposition of an array using the tool `numpy.transpose`.

It will not affect the original array, but it will create a new array.

```
import numpy

my_array = numpy.array([[1,2,3],
                        [4,5,6]])
print numpy.transpose(my_array)
```

```
#Output
[[1 4]
 [2 5]
 [3 6]]
```

## Flatten

The tool `flatten` creates a copy of the input array flattened to one dimension.

```
import numpy

my_array = numpy.array([[1,2,3],
                        [4,5,6]])
print my_array.flatten()
```

```
#Output
[1 2 3 4 5 6]
```

Author [\[deleted\]](#)

Difficulty [Easy](#)

Max Score 20

Submitted By [128635](#)

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## Task

You are given a  $N \times M$  integer array matrix with space separated elements ( $N$  = rows and  $M$  = columns).

Your task is to print the transpose and flatten results.

## Input Format

The first line contains the space separated values of  $N$  and  $M$ .

The next  $N$  lines contains the space separated elements of  $M$  columns.

## Output Format

First, print the transpose array and then print the flatten.

## Sample Input

```
2 2
1 2
3 4
```

## Sample Output

```
[[1 3]
 [2 4]]
[1 2 3 4]
```

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Language

Pypy 3



```
1
2
3
4
5
6
7
```

Line: 7 Col: 21



Upload Code as File



Test against custom input

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